

ECON-1020H Introductory Macroeconomics

Chapter 7 Production and Growth

Shutao Cao

2025-09-29

Trent University
Fall 2025

Outline

What is growth?

Economic Growth: Basic Picture

Growth Rates: Rule of 70

Sources of economic growth

Growth of Output per Worker

Disparity in economic growth

Other Topics of Growth

Recap

What is growth?

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Other Topics of Growth

Recap

Reading of the week:

- Textbook Chapter 7

Reading of next week:

- Textbook Chapter 8

What is growth?

Economic growth defined

Economic growth refers to sustained increases in aggregate output and improvements in the standard of living

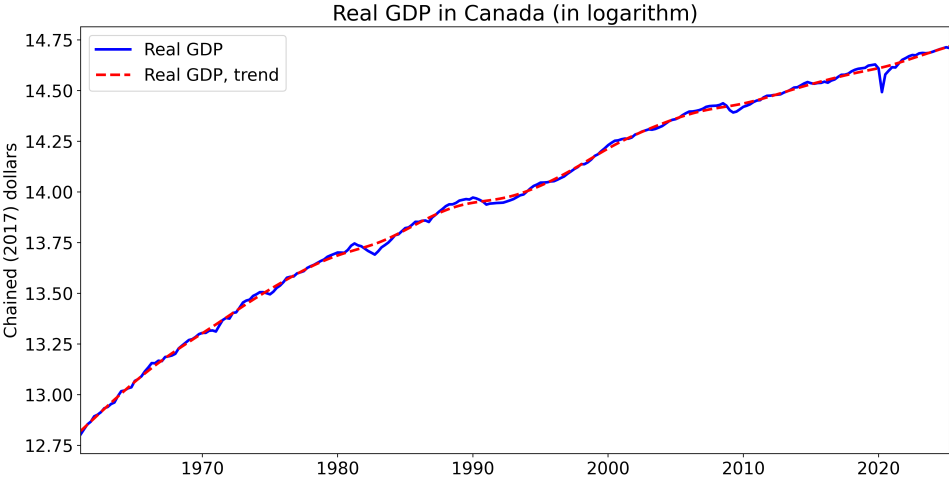
The standard of living is measured as **real GDP per capita**—the value of real GDP divided by population size

- It is real income per person, also real expenditure per person
- Often used to track economic growth and for cross-country comparison

This chapter studies drivers of long-run economic growth

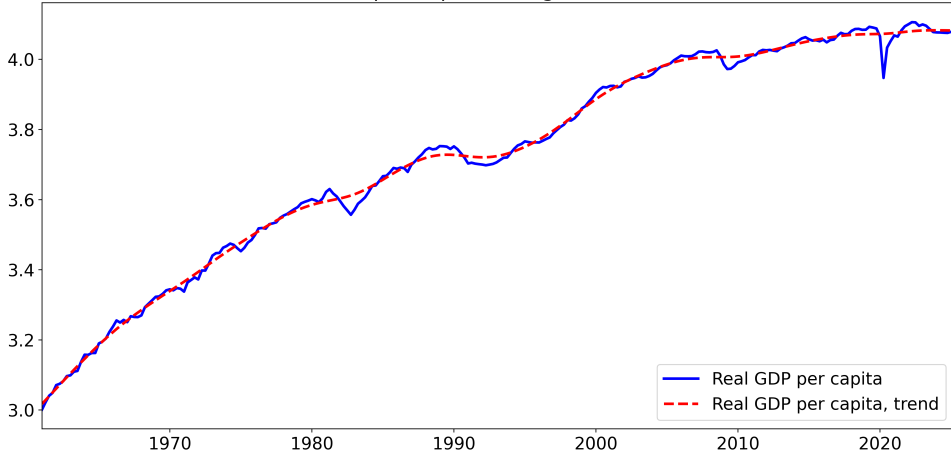
Economic Growth: Basic Picture

Economic growth in Canada



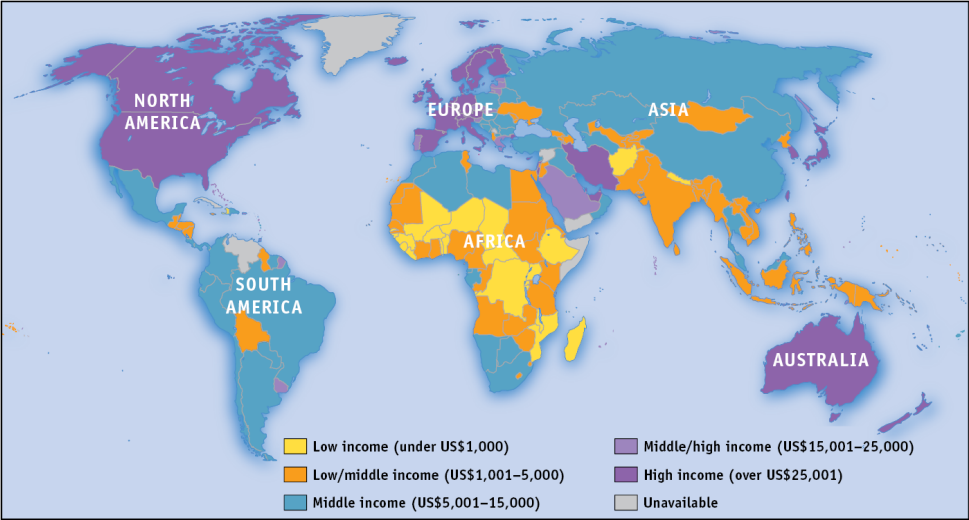
Data source: Statistics Canada

Real GDP per capita (in logarithm), Canada



Data source: Statistics Canada

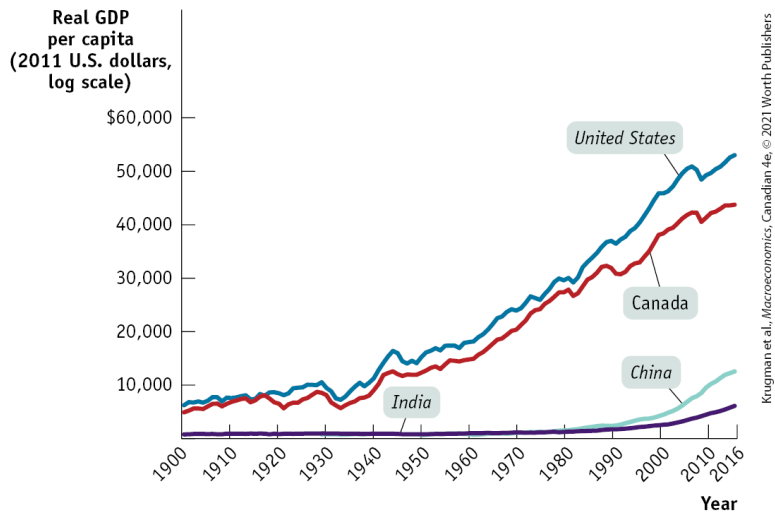
Across countries: income differences, 2018



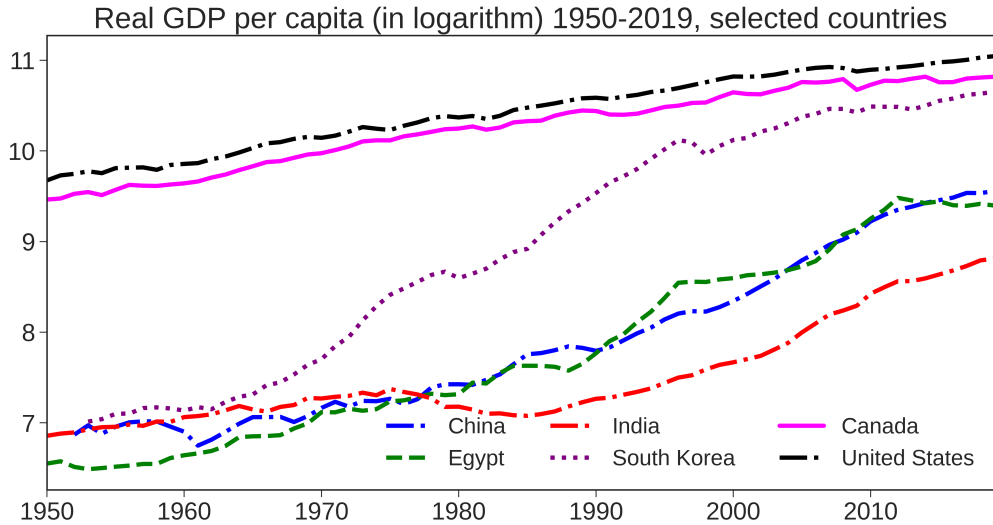
Krugman et al., *Macroeconomics*, Canadian 4e, © 2021 Worth Publishers

Across countries: a very long run

Economic growth is a modern phenomenon, after the 1st industrial revolution

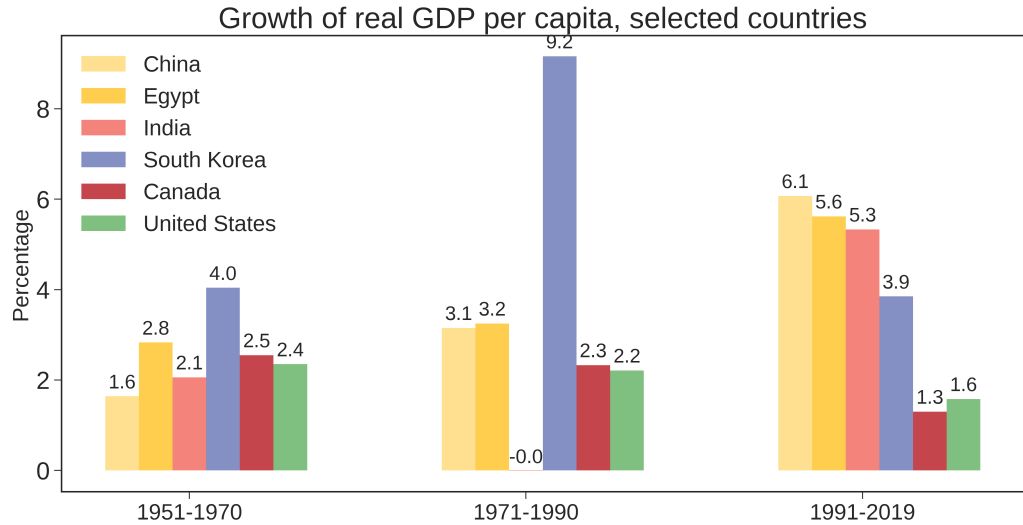


Across countries: past 70 years



Data source: Penn World Table v10.01

Across countries: past 70 years



Data source: Penn World Table v10.01

What is striking to Robert Lucas Jr

To quote Robert E. Lucas Jr. (1988):

Is there some action a government of India could take that would lead the Indian economy to grow like Indonesia's or Egypt's? If so, what, exactly? If not, what is it about the "nature of India" that makes it so? The consequences for human welfare involved in questions like these are simply staggering: Once one starts to think about them, it is hard to think about anything else

Growth Rates: Rule of 70

Growth rates: Rule of 70

Growth rate of a variable is its percentage change over time

How long does it take for real GDP per capita (or any variable) to **double**?

The Rule of 70

The number of years for a variable to double

$$= \frac{70}{\text{annual growth rate of the variable}}$$

Rule of 70 applied to Canada

- In the past seven decades or so, real GDP per capita in Canada rose on average 2% per year
- By the rule of 70, it takes 35 ($=70/2$) years for real GDP per capita to double in Canada
- In recent two decades, growth of real GDP per capita slowed down
- If real GDP per capita grows 1.5% per year, how long does it take for real GDP per capita to double?
 - 47 years

Practice problem 1

1. Real GDP per capita in China, India, and Egypt grew at about 6 percent per year since 1990, how many years does it take for real GDP per capita to double in these countries?
2. Real GDP per capita in South Korea grew at 9% per year during 1971-1990. Suppose, in 1971, real GDP per capita in south Korea was 2980 dollars in real price. What was the level of real GDP per capita in 1990?

(Answer)

Sources of economic growth

Sources of growth in real GDP per capita

$$\text{Real GDP per capita} = \frac{\text{Real GDP}}{\text{Population}} = \frac{\text{Real GDP}}{\text{Employment}} \times \frac{\text{Employment}}{\text{Population}}$$

Growth rate of real GDP per capita =

$$\text{Growth rate of } \left(\frac{\text{Real GDP}}{\text{Employment}} \right) + \text{Growth rate of } \left(\frac{\text{Employment}}{\text{Population}} \right).$$

$\frac{\text{Real GDP}}{\text{Employment}}$ is called **output per worker** or **real GDP per worker**

- Textbook calls it "Productivity"
- In fact, in textbook, productivity is defined as real GDP per hour worked, but then later textbook confuses readers by using "real GDP per worker" for productivity

Sources of growth of real GDP per capita

Growth rate of real GDP per capita = growth rate of output per worker + growth rate of employment to population ratio

Employment-population ratio changes little in the long run

- It accounts for less than 1/5 of growth of real GDP per capita

Growth of real GDP per capita is primarily driven by output per worker

Then, we should focus on the growth of output per worker

Growth of Output per Worker

Aggregate production

In order to study the growth of output per worker, we should think about how output is produced (determined)

In the long run, real GDP is mainly determined by the available production technology and factors of production

Why? In a free-market economy, it is unlikely that factors of production are wasted or idle for a prolonged period of time

We can imagine that aggregate output is produced with a production function involving technology and factors of production

Let Y be aggregate output (real GDP), aggregate production function is

$$Y = A \cdot F(K, L, H)$$

- A is the total factor productivity (TFP)
- K is the stock of capital
- L is the amount of labor input (can think of it as employment)
- H is the stock of human capital

Intuition: the economy uses capital, labor, knowledge (human capital), and some technology to produce output

- Y increases if any of K , L , and H increases, or all three inputs increase

Why are intermediate inputs absent in the production function?

- GDP is the aggregate value added, which is also total factor income

Total factor productivity

- TFP represents factors that increase output without needing to increase the use of factors of production (K, L, H)

Some of things causing TFP to change:

- research and development
- management practice
- institutions
- government regulation

Output per Worker

Output per worker is expressed as

$$\frac{Y}{L} = A \cdot \frac{F(K, L, H)}{L}.$$

Let $\frac{F(K, L, H)}{L} = f(K/L, H/L)$, output per worker becomes

$$\frac{Y}{L} = A \cdot f(K/L, H/L)$$

- The right-hand side is called **per-worker production function**
- $\frac{K}{L}$ is capital-labor ratio
- $\frac{H}{L}$ is human capital per worker, a measure of average labor quality

FYI: Transforming from function $F()/L$ to $f()$ needs an assumption, beyond this course

Sources of Growth in Output per Worker

The per-worker production function $\frac{Y}{L} = A \cdot f(K/L, H/L)$ suggests:

Growth of output per worker arises from

- technological progresses (TFP growth)
- growth of physical capital per worker
- growth of human capital per worker

Example: $\frac{Y}{L} = A \cdot (K/L)^{0.4} \cdot (H/L)^{0.6}$

- Exponents 0.4 and 0.6 are consistent with some data
- Growth rate of $Y/L =$
growth rate of $A + 0.4 \times$ growth rate of $K/L + 0.6 \times$ growth rate of H/L

Growth of physical capital per worker

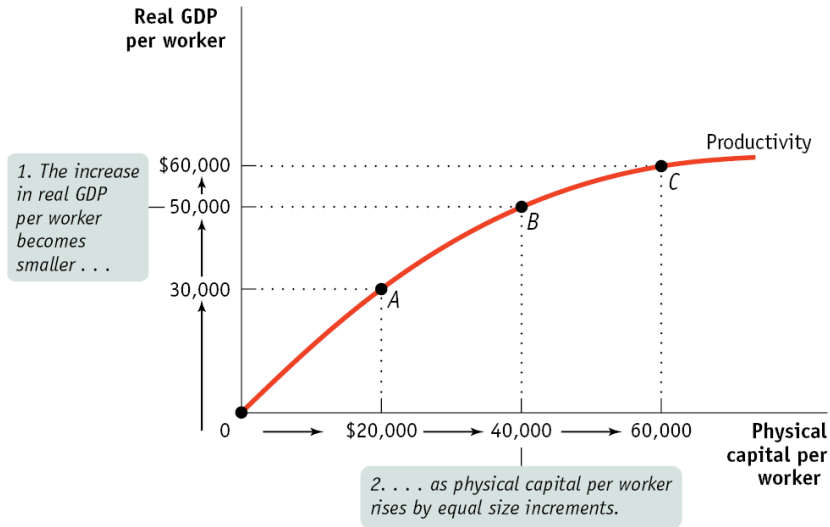
The growth of physical capital per worker leads to growth of output per worker

However, such effect dwindles as physical capital per worker becomes higher

Why? **Diminishing marginal product of capital**

Diminishing marginal product of physical capital

Holding A and H/L constant, each successive increase in the amount of physical capital per worker leads to a smaller increase in output per worker, as physical capital per worker becomes higher



Physical capital per worker	Real GDP per worker
\$ 0	\$ 0
20,000	30,000
40,000	50,000
60,000	60,000

Catch-up effect

The diminishing marginal product of capital per worker implies the catch-up effect:

- Income level in countries with low capital per worker will catch up income in countries with high capital per worker
- This is because an increase in capital per worker improves output more when capital per worker is low

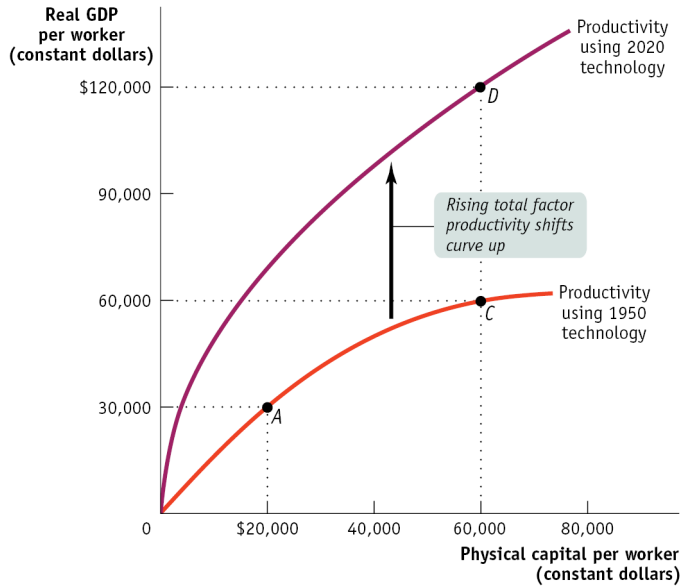
It is also called the "convergence"

- There is evidence supporting convergence among OECD countries
- But, empirically, convergence has not happened globally

When TFP rises,

- the economy produces more even if K/L and H/L do not change,
- leading to a higher output per worker

NOTE: A higher TFP raises the marginal output-per-worker of K/L , see next graph



Growth accounting is an exercise of decomposing the growth of output per worker into contributions from different factors

Growth accounting for output per worker:

- Contribution from increases in TFP (technological progress)
- Contribution from the growth of capital per worker (K/L)
- Contribution from the growth of labor quality (H/L)

Let growth rates be $g_{Y/L}$, g_A , $g_{K/L}$, $g_{H/L}$

Growth accounting of labor productivity

Suppose $\frac{Y}{L} = A \cdot (K/L)^{0.4} \cdot (H/L)^{0.6}$

Then, $g_{Y/L} = g_A + 0.4 \cdot g_{K/L} + 0.6 \cdot g_{H/L}$

Note: The exponents do not have to be 0.4 and 0.6, but two exponents add to one

A numerical example

Continue using the function for Y/L as in previous slide

Suppose capital per worker grows 3% per year, human capital per worker grows 0.5% per year, and labor productivity grows 3% per year

What is the contribution of TFP to labor productivity growth?

$$\underbrace{0.03}_{g_{Y/L}} = \underbrace{\text{growth rate of } A}_{g_A} + 0.4 \cdot \underbrace{0.03}_{g_{K/L}} + 0.6 \cdot \underbrace{0.005}_{g_{H/L}}.$$

$$\text{TFP growth rate } g_A = 0.03 - 0.4 \cdot 0.03 - 0.6 \cdot 0.005 = 0.015 = 1.5\%$$

Table 1: Average annual growth rates (percent), business sector

Period	1961-2023	1961-2000	2000-2023	2019-2023
Real GDP per worker:	1.73	2.29	0.93	0.30
- Contribution of K/L	0.93	1.08	0.70	0.40
- Contribution of H/L	0.38	0.45	0.27	0.29
- TFP	0.42	0.76	-0.04	-0.39

For the entire economy, from 2019q1 to 2025q2 on average:

- Real GDP per capita declined 0.137 percent annually
- Employment-population ratio declined 0.059 percent annually
- Real GDP per employment declined 0.078 percent annually

What about natural resources?

- Historically, they are less important than technological progress and human capital
- Look into the future, will natural resources exhaust? Yes/No, beyond this course

Population Growth and Output per Worker

In the per-worker production function, $Y/L = Af(K/L, h/L)$, population growth plays little role

English economist Malthus thought that real GDP per capita would fall as population grows, because land is fixed

- In Malthus' model, land is the only physical capital, and has a fixed supply

Malthus' concern was proven to be wrong during his own time (1800s)

- He ignored both technological progress,
- and growth of machinery capital

Still, the socialism Chinese government adopted Malthus' theory

- It implemented one-child policy from 1979 to 2015

Disparity in economic growth

What explain cross-country differences in growth rates?

Both levels and growth rates of the standard of living differ across countries

Per-worker production function suggests, high growth rate results from

- High saving and investment spending, leading to growth in K/L
- Improvements in educational attainment, leading to growth in H/L
- More spending on Research and Development, leading to growth in TFP

Governments have made themselves "indispensable" in economic growth

- Some governments invest heavily in infrastructure, others subsidize education and R&D
- Protection of property rights
- Political stability and sound governance, hard to quantify, but crucial

Nowadays, Milton Friedman appears unpopular

Other Topics of Growth

Real GDP per capita says little about income distribution

Real GDP growth benefits people differently

Recent two trends:

1. upward trends in income inequality;
2. slower real GDP growth

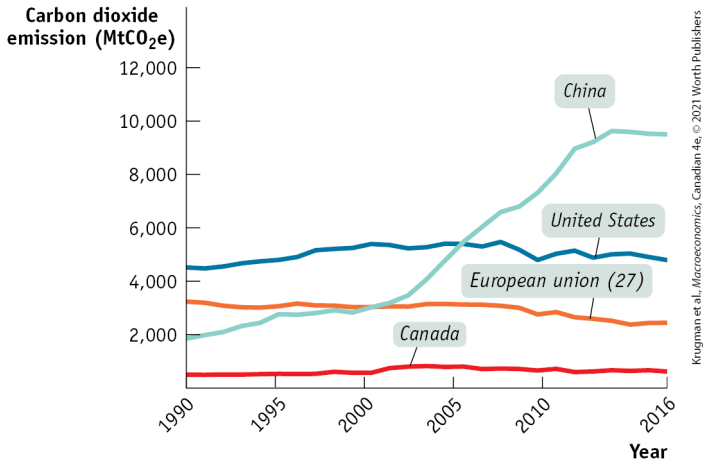
Does increased inequality hinder growth?

- Not conclusive among economists
- Kuznets curve: Income inequality rises as economy grows, after some stage of growth, income inequality starts to shrink
- Kuznets curve is hypothesis, it fails in recent trends

Growth and the Environment

The most imminent issue is probably the climate change

- Is it a zero-sum?



According to [EIA.GOV](https://www.eia.gov), energy-intensive productions:

- food, pulp and paper, basic chemicals, refining, iron and steel, nonferrous metals (primarily aluminum), and nonmetallic minerals (primarily cement)

In the U.S., the most energy intensive industry is cement production

How should the world economy reduce carbon dioxide emission?

- Command-and-control versus market-based mechanisms
- There exists no global authority that can make orders
- Is climate change the "tragedy of commons"?

Recap

Recap

- Growth facts over time and across countries
- The rule of 70
- Sources of the growth of real GDP per capita
- Sources of growth in output per worker

Answers to practice problem 1

(1) It takes 12 years

(2) Real GDP per capita in 1990 was about 4.5 times that in 1971

- Real GDP per capita in 1990 is 2980 times 4.5
- It takes 7.78 years for real GDP per capita to double
- Then, it takes 15.555 years for that to quadruple
- And, it takes $(7.78/2) = 3.889$ years for real GDP per capita to increase 50%
- Therefore, in $(15.555+3.889) = 19.44$ years, real GDP per capita became 4.5 folds of that in 1971

[\(Go back to problems\)](#)